

Lecture 4: Functions of Random Variables, Linearity, Variance

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1 A linear function of a random variable

Let C be a random variable — today's high temperature in Celsius, say, with some PMF p_C . In Fahrenheit,

$$F = g(C) = \frac{9}{5}C + 32.$$

F is a new random variable on the *same* sample space as C , defined by $F(\omega) = g(C(\omega))$.

2 The PMF of a transformed RV

If g is **one-to-one** (invertible) on the range of X , the PMF of $Y = g(X)$ is easy:

$$p_Y(y) = P(Y = y) = P(g(X) = y) = P(X = g^{-1}(y)) = p_X(g^{-1}(y)).$$

For Fahrenheit, $g^{-1}(f) = \frac{5}{9}(f - 32)$, so $p_F(f) = p_C(\frac{5}{9}(f - 32))$ — the probabilities don't change, only the labels on the axis do.

3 LOTUS: the Law Of The Unconscious Statistician

What if we only want $E[Y]$ for $Y = g(X)$, and g *isn't* one-to-one — so finding p_Y takes real work (recall grouping outcomes by value, as we did for the dice-game payout in Lecture 3)?

i Theorem: LOTUS

For a discrete RV X and any function g ,

$$E[g(X)] = \sum_x g(x) p_X(x).$$

The name is a joke: it feels too good to be true, but it is!

4 Distributions

i Definition: Distribution

The **distribution** of a random variable X describes how the total probability (which always sums to 1) is spread out across the values X can take. Formally, it's specified by giving $P(X \in B)$ for every relevant subset B of values that X can take (or more generally, every subset $B \subset \mathbb{R}$).

For a **discrete** RV, the distribution is fully captured by the PMF $p_X(x) = P(X = x)$ — knowing p_X at every possible value tells you $P(X \in B)$ for any B , just by summing. (For RVs with uncountably many values, a PMF alone won't do — we'll need new tools, coming in a later lecture.)

5 LOTUS example 1: the dice-game payout, revisited

Recall X = die roll (uniform on $\{1, \dots, 6\}$), and $Y = g(X)$ pays \$2 on even, -\$1 on odd. In Lecture 3 we found p_Y first, then computed $E[Y]$. LOTUS skips straight to the answer:

$$E[Y] = \sum_{x=1}^6 g(x) \cdot \frac{1}{6} = -\frac{1}{6} + \frac{2}{6} - \frac{1}{6} + \frac{2}{6} - \frac{1}{6} + \frac{2}{6} = \frac{1}{2}.$$

6 LOTUS example 2: randomized Kaczmarz, revisited

Recall randomized Kaczmarz (RK), solving $Ax = b$: at each step we sample a row index $I \in \{1, \dots, m\}$ at random and update

$$x_{k+1} = x_k + \frac{b_I - a_I^\top x_k}{\|a_I\|^2} a_I.$$

A popular distribution over the rows is the **row-norm PMF** $p_I(i) = \|a_i\|^2 / \|A\|_F^2$. We'll see why now!

Fix the current iterate x_k and let $v = x_k - x^*$. LOTUS lets us compute the expected value of *any* function of I straight from p_I — for instance

$$E\left[\frac{(a_I^\top v)^2}{\|a_I\|^2}\right] = \sum_{i=1}^m \frac{(a_i^\top v)^2}{\|a_i\|^2} p_I(i).$$

Plugging in $p_I(i) = \|a_i\|^2 / \|A\|_F^2$, the $\|a_i\|^2$ cancels:

$$\sum_{i=1}^m \frac{(a_i^\top v)^2}{\|a_i\|^2} \cdot \frac{\|a_i\|^2}{\|A\|_F^2} = \frac{1}{\|A\|_F^2} \sum_{i=1}^m (a_i^\top v)^2 = \frac{\|Av\|^2}{\|A\|_F^2}.$$

If I were sampled **uniformly** instead, the $\|a_i\|^2$ would *not* cancel, and the final form wouldn't be so nice and simple!

7 Linearity of expectation

i Theorem: Linearity of expectation

For random variables X, Y on the same probability space — **possibly dependent** — and constants a, b, c :

$$E[aX + bY + c] = aE[X] + bE[Y] + c.$$

⚠ No independence required!

This holds *no matter how X and Y are related*. It's what makes linearity of expectation so powerful. (Variance will not be so forgiving — stay tuned.)

8 The mean of a Binomial, the slick way

Recall from Lecture 3 we derived $E[X] = np$ for $X \sim \text{Binomial}(n, p)$ via a combinatorial identity. Linearity gives a one-line proof instead.

i Definition: Indicator random variable

For an event A , the indicator $\mathbf{1}_A$ equals 1 if A occurs, 0 otherwise. Note $E[\mathbf{1}_A] = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A)$.

i Definition: i.i.d.

Random variables $X_1, \dots, X_n$ are **independent and identically distributed (i.i.d.)** if they are mutually independent and all share the same distribution.

Write binomial random variable $X = \sum_{i=1}^n X_i$, where $X_i = \mathbf{1}_{\{\text{trial } i \text{ succeeds}\}}$ — i.i.d. Bernoulli(p) indicators. By linearity (independence isn't even needed here!):

$$E[X] = \sum_{i=1}^n E[X_i] = \sum_{i=1}^n p = np.$$

9 Linearity also applies to functions of X and Y

Combining LOTUS with linearity, for any functions g, h :

$$E[g(X) + h(Y)] = E[g(X)] + E[h(Y)].$$

We'll use exactly this form below.

10 The coupon collector problem

There are n distinct coupon types; each box of cereal contains one coupon, uniformly random among the n types, independent of previous boxes. Let T = number of boxes you need to buy in order to collect all n types of coupons. What's $E[T]$?

Write $T = T_1 + T_2 + \dots + T_n$, where T_i = number of *additional* boxes needed to find a new type, right after collecting the $(i-1)$ -th distinct type. With $i-1$ types already in hand, each new box is a new type with probability $p_i = \frac{n-i+1}{n}$, so $T_i \sim \text{Geometric}(p_i)$ and $E[T_i] = \frac{n}{n-i+1}$.

By linearity,

$$E[T] = \sum_{i=1}^n E[T_i] = \sum_{i=1}^n \frac{n}{n-i+1} = n \sum_{k=1}^n \frac{1}{k} = nH_n \approx n \ln n.$$

```
import numpy as np

def collect_coupons(n, rng):
    seen = set()
    draws = 0
    while len(seen) < n:
        seen.add(rng.integers(n))
        draws += 1
    return draws

rng = np.random.default_rng(151)
n = 20
trials = 3000
results = [collect_coupons(n, rng) for _ in range(trials)]

H_n = sum(1 / k for k in range(1, n + 1))
print(f"n = {n}")
print(f"simulated average T over {trials} trials: {np.mean(results):.2f}")
print(f"formula n*H_n = {n*H_n:.2f}")
```

```

n = 20
simulated average T over 3000 trials: 72.50
formula n*H_n = 71.95

```

11 Linearity + LOTUS: average one-step progress of RK

Recall randomized Kaczmarz's update, and note that the Pythagorean theorem (the update is an orthogonal projection) provides,

$$\|x_{k+1} - x^*\|^2 = \|v\|^2 - \frac{(a_I^\top v)^2}{\|a_I\|^2}.$$

where $v = x_k - x^*$.

$\|v\|^2$ is a **constant** given x_k — only I is random. By linearity,

$$E[\|x_{k+1} - x^*\|^2] = \|v\|^2 - E\left[\frac{(a_I^\top v)^2}{\|a_I\|^2}\right] = \|v\|^2 - \frac{\|Av\|^2}{\|A\|_F^2},$$

reusing the LOTUS computation from a few slides ago!

Since $\|Av\| \geq \sigma_{\min}(A)\|v\|$,

$$E[\|x_{k+1} - x^*\|^2] \leq \left(1 - \frac{\sigma_{\min}(A)^2}{\|A\|_F^2}\right) \|x_k - x^*\|^2.$$

Every step shrinks the *expected* squared error by a fixed factor — this a very nice (and very well-cited) result from Thomas Strohmer and Roman Vershynin in 2009!

12 Watching the bound play out over many steps

The one-step bound says the expected squared error shrinks by a fixed factor *every* step. Let's actually run RK and watch $\|x_k - x^*\|^2$ fall, averaged over many independent random runs.

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(151)
m, n = 20, 8
A = rng.standard_normal((m, n))
x_star = rng.standard_normal(n)
b = A @ x_star
row_norm_sq = (A ** 2).sum(axis=1)
p_row_norm = row_norm_sq / row_norm_sq.sum()
sigma_min = np.linalg.svd(A, compute_uv=False)[-1]
frob_sq = row_norm_sq.sum()

num_steps = 60
num_runs = 300
errors_sq = np.zeros((num_runs, num_steps + 1))

for r in range(num_runs):
    x = np.zeros(n)
    errors_sq[r, 0] = np.linalg.norm(x - x_star) ** 2
    idx_sequence = rng.choice(m, size=num_steps, p=p_row_norm)

```

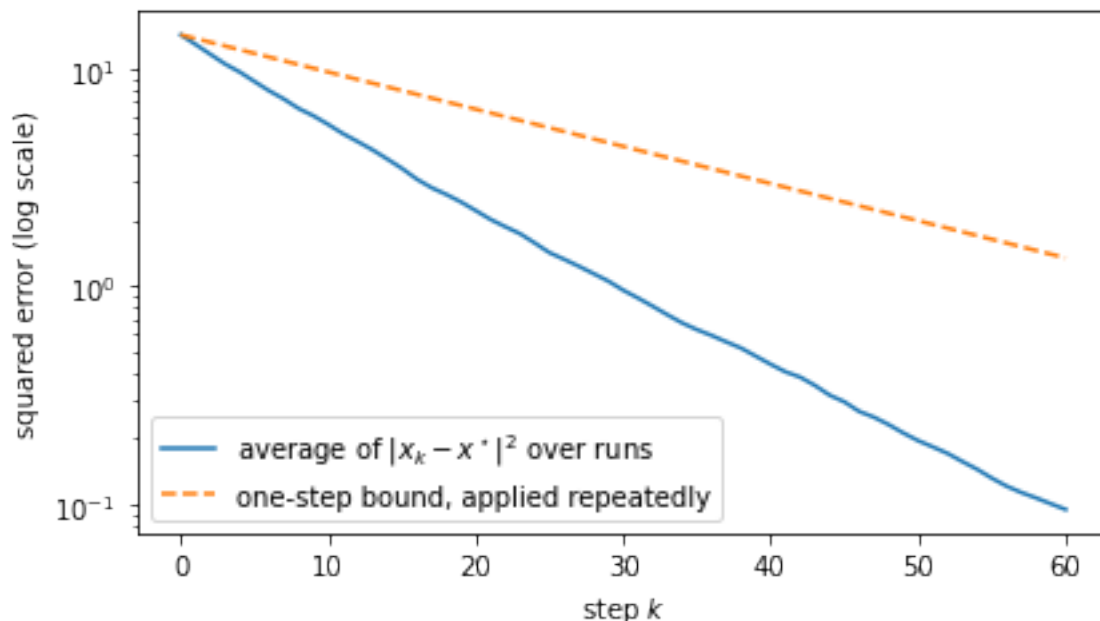
```

for t, i in enumerate(idx_sequence, start=1):
    a_i, b_i = A[i], b[i]
    x = x + (b_i - a_i @ x) / row_norm_sq[i] * a_i
    errors_sq[r, t] = np.linalg.norm(x - x_star) ** 2

mean_error_sq = errors_sq.mean(axis=0)
bound_curve = errors_sq[:, 0].mean() * (1 - sigma_min**2 / frob_sq) ** np.arange(num_steps + 1)

fig, ax = plt.subplots(figsize=(6, 3.5))
ax.semilogy(mean_error_sq, label="average of  $\|x_k - x^*\|^2$  over runs")
ax.semilogy(bound_curve, "--", label="one-step bound, applied repeatedly")
ax.set_xlabel("step  $k$ ")
ax.set_ylabel("squared error (log scale)")
ax.legend()
plt.tight_layout()
plt.show()

```



13 Moments

i Definition: Moments

The k -th moment of a random variable X is $E[X^k]$. The mean $E[X]$ is the 1st moment. The 2nd moment, $E[X^2]$, is about to become useful.

14 Why do we need more than the mean?

Suppose you only know $E[X]$ where $X = \|x_k - x^*\|^2$ is small (e.g., from the bound we just derived). What does that tell you about the error on *any one particular run* of RK iterations?

Not much, on its own! The mean is an average over all randomness; a single run could still land far from it. To say anything about how *spread out* outcomes are around the mean, we need a new tool: **variance**.

💡 Aside: the probabilistic method

Expectation alone *can* still be a powerful existence tool: if $E[X] \leq c$, then *some* outcome ω must satisfy $X(\omega) \leq c$ (otherwise every outcome would exceed c , forcing $E[X] > c$). This “probabilistic method” is often used to prove existence results in combinatorics — but it says nothing about *how many* outcomes are near c , or how far the rest can stray. That’s what variance is for.

15 Variance and standard deviation

i Definition: Variance

$$\text{Var}(X) = E[(X - E[X])^2].$$

The **standard deviation** is $\sigma_X = \sqrt{\text{Var}(X)}$ — same units as X itself (variance is in squared units), which is often why it’s the more interpretable number.

16 Variance of a linear function

For constants a, b :

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

Proof: $E[aX + b] = aE[X] + b$, so

$$\text{Var}(aX + b) = E[(aX + b - aE[X] - b)^2] = E[a^2(X - E[X])^2] = a^2 \text{Var}(X).$$

The additive shift b vanishes entirely — variance sees only *spread*, never *location*.

17 Variance shortcut formula

i Theorem

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$

Proof: expand $(X - E[X])^2 = X^2 - 2XE[X] + (E[X])^2$ and take $E[\cdot]$, using linearity (note $E[X]$ is a constant):

$$E[(X - E[X])^2] = E[X^2] - 2E[X]E[X] + (E[X])^2 = E[X^2] - (E[X])^2.$$

This is almost always the easiest way to *compute* a variance.

18 Compute $\text{Var}(X)$ for each named distribution

Bernoulli(p): $X^2 = X$ on $\{0, 1\}$, so $E[X^2] = E[X] = p$, giving $\text{Var}(X) = p - p^2 = p(1 - p)$.

Rademacher: $X^2 = 1$ always, so $E[X^2] = 1$, $E[X] = 0$, giving $\text{Var}(X) = 1$.

🔥 In-class activity

Using $\text{Var}(X) = E[X^2] - (E[X])^2$, compute $\text{Var}(X)$ for Geometric and Poisson.

Geometric(p): Using the derivative series trick twice gives $E[X^2] = \frac{2-p}{p^2}$, so

$$\text{Var}(X) = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}.$$

Poisson(λ): $E[X^2] = \lambda^2 + \lambda$, which gives

$$\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

(Poisson is the rare distribution with mean = variance!)

19 Variance of a sum — independence *is* required this time

 Contrast with linearity of expectation

$E[X + Y] = E[X] + E[Y]$ *always*. But in general $\text{Var}(X + Y) \neq \text{Var}(X) + \text{Var}(Y)$ — cross terms from how X and Y move together get in the way.

 Theorem

If $X_1, \dots, X_n$ are **independent**, then

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

Payoff: writing $\text{Binomial}(n, p)$ as a sum of n i.i.d. $\text{Bernoulli}(p)$ indicators:

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) = np(1-p).$$

20 Central moments

 Definition: Central moments

The k -th **central moment** of X is $E[(X - E[X])^k]$ — an ordinary moment, but centered at the mean first.

- The 1st central moment is always 0 (by definition of the mean).
- The **2nd central moment is exactly the variance** — nothing new, just a name.
- Higher central moments capture other shape features: the 3rd relates to *skewness* (asymmetry), the 4th to *kurtosis* (tail heaviness). We won't need these in this course, but variance is really just the first member of this whole family.